

OT GoalAssist

Feasibility Presentation: Iteration 1

"Helping Occupational Therapists COAST Through the Day."

Course:	CS 410: Professional Workforce Development (Fall 2026)
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Team Iron: Project Roles & Responsibilities

Team Member	Project Role	Primary Responsibilities & Core Deliverables
Hope Chandler	**Team lead / Project author**	Original concept author, clinical requirements, OT workflow, COAST rules, and assessment criteria.
Max Adams	**Webmaster / UI**	Repository maintenance, GitHub Pages website developer and UI.
Hamza Guzel	**Front-end developer**	Developing and maintaining the front-end interface, including assessment and patient input forms, COAST goal review screens, missing-information alerts, and therapist sign-off.
Monte Jacob	**Back-end developer**	Developing and maintaining back-end components, implementing system logic, managing data, and developing the functionalities of the system.
Mohamed Datt	**Database specialist**	Relational data schema, entity relationship diagrams (ERD), assessment score storage, and goal history.
Austin Jones	**Full-stack developer**	Development of both front-end and back-end systems, coordination between these systems, and holistic project evaluation.
Trevor Prince	**Documentation specialist**	Maintenance of accurate documentation of the project as it progresses including its prerequisites, design changes, use-cases, and overall architecture.

What is Occupational Therapy?

- Occupational therapy (OT) is a critical and fast-growing healthcare field that directly improves patient independence, daily functioning, and quality of life
- Occupational therapists (OTs) work with patients from the NICU to hospice care and in many healthcare settings and social systems including schools and prisons, adding holistic and client centered perspective on interdisciplinary healthcare teams
- More people are living with a disability now compared to ever before as technological advancements have increased the ability to live with a disability.

6 million

OTs provide approximately 6 million life-changing OT sessions every year

MarketingInsights. (2024, July 16). *How occupational therapy changes people's lives - ability rehabilitation*. Ability Rehabilitation. https://abilityrehabilitation.com/news/how-occupational-therapy-changes-peoples-lives/

15% growth

OTs is projected to grow 15% from 2025 to 2035, adding roughly 10,000 annual openings

U.S. BUREAU OF LABOR STATISTICS. (2019, April 12). *Occupational therapists : Occupational outlook handbook: : U.S. Bureau of Labor Statistics*. Bls.Gov; U.S. Bureau of Labor Statistics. https://www.bls.gov/ooh/healthcare/occupational-therapists.htm

$92.38 billion

Occupational and physical therapy is projected to grow from $53.08 billion in 2023 to $92.38 billion by 2030

US Occupational & Physical Therapy Services market size to surpass USD 92.38 billion By 2030, at a CAGR of 8.2%. (2023, October 10). *YAHOO!Finance*. https://finance.yahoo.com/news/us-occupational-physical-therapy-services-125500385.html

The Problem Domain: Occupational Therapy in Healthcare

What Occupational Therapists Do	The Administrative Disconnect
• Functional Rehabilitation Focus: Occupational therapists help patients of all ages develop, recover, or maintain essential Activities of Daily Living (ADLs) such as dressing, eating, hygiene, and fine motor writing. • Individualized Plans of Care: Every client requires an individualized Plan of Care (POC) with explicit, measurable functional goals tailored to their unique physical or cognitive deficits. • Standardized Objective Testing: Clinicians administer comprehensive assessment batteries including the BOT-2 (Bruininks-Oseretsky Test of Motor Proficiency), PDMS-2 (Peabody Developmental Motor Scales), and MMT (Manual Muscle Testing).	• Rigorous Insurance Reimbursement Mandates: Payers (Medicare, Medicaid, private insurers) demand objective, standardized proof of progress before approving billable therapy sessions. • Manual Data Translation Bottleneck: Therapists must manually calculate percentiles and bridge raw numerical scores into structured, compliant clinical prose from scratch for each patient. • Diverse and Fragmented Practice Settings: Clinicians practice across diverse settings (outpatient pediatric clinics, acute hospital rehab, school districts), each demanding specialized documentation criteria.

The Documentation Burden: Scale, Pressure, and Burnout

75% - 85%	70%+	48%
Mandated Productivity Quotas	**Unpaid "Pajama Time"**	**Clinician Burnout Rate**
Healthcare clinics require therapists to spend 75% to 85% of their working hours on direct, billable patient treatment, leaving under 15% for all documentation and care planning.	Over 70% of allied healthcare clinicians report bringing documentation home to finish at night off the clock because clinical hours are fully consumed by patient sessions.	48% of surveyed occupational therapists cite excessive unbillable paperwork as the primary driver of emotional exhaustion, moral injury, and career turnover.

Key Industry Evidence:
• American Occupational Therapy Association (AOTA) Workforce & Practice Trends Survey (2023).
• Medscape Clinician Burnout & Depression Report (2024).
When therapists are forced to spend hours manually typing care plans, direct patient care quality declines, clinical fatigue spikes, and healthcare systems suffer costly staff turnover.

The Clinical Goal Standard: The COAST Framework

Occupational therapists use the COAST framework (Gateley & Borcherding) to guarantee that patient goals are specific, measurable, and reimbursable:

Part	Component	Clinical Definition & Operational Rule	Clinical Example Phrase
C	**Client**	Identifies who performs the functional action.	*"Client will..."*
O	**Occupation**	Specifies the meaningful daily living task or occupation.	*"...fasten 4 standard shirt buttons..."*
A	**Assist Level**	Quantifies the exact physical or verbal assistance required.	*"...with minimal physical assistance..."*
S	**Specific Conditions**	Defines adaptive equipment, location, or cueing parameters.	*"...using a button-hook tool while seated..."*
T	**Time Frame**	Establishes a clear, measurable achievement deadline.	*"...within 4 weeks (by discharge)."*

The Clinical Bottleneck: Missing just one component (such as an explicit assistance level or condition) triggers insurance claim rejections. Manually verifying all five parameters across multiple goals per patient is repetitive and cognitively exhausting.

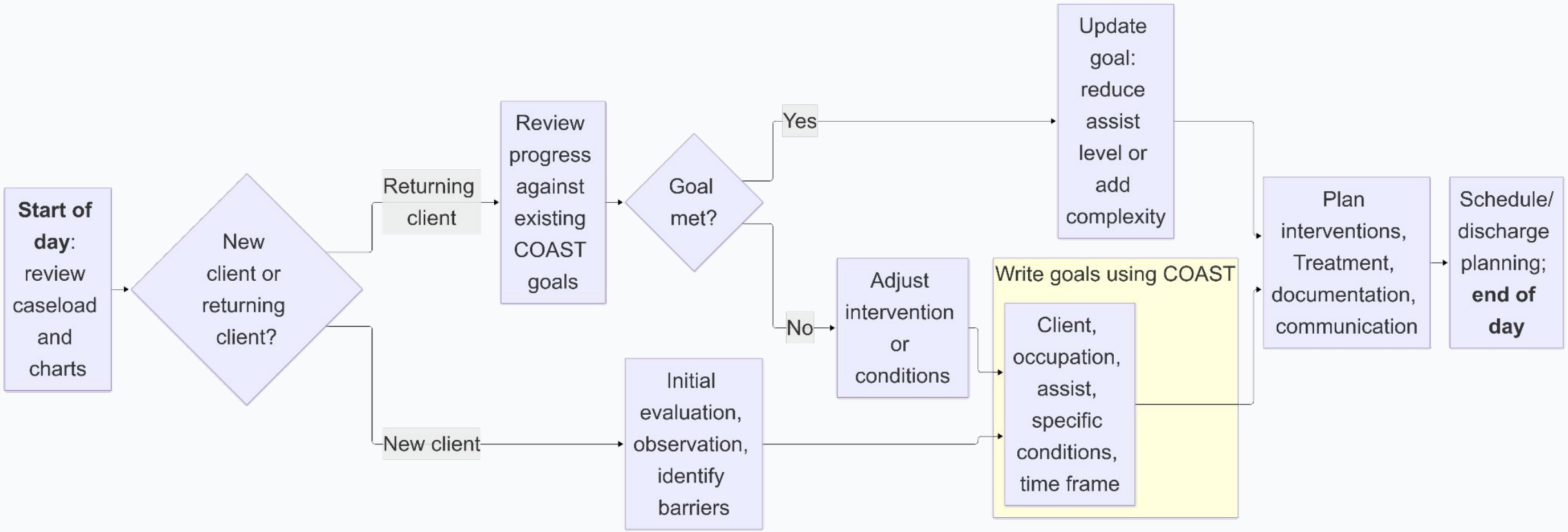
Target Ecosystem: Users, Customers, and Stakeholders

Direct Users	Customers	Stakeholders
Clinicians using the tool daily	*Organizations purchasing software*	*Beneficiaries & regulatory bodies*
• Registered Occupational Therapists (OTRs) • Certified Occupational Therapy Assistants (COTAs) • Core Pain Points: Lost personal evenings to paperwork ("pajama time"), mental fatigue from synthesizing test batteries, fear of insurance audits.	• Outpatient Therapy Private Practices • Hospital Rehabilitation Networks • School Districts & Pediatric Centers • Business Value: Maximizes billable treatment compliance, reduces clinician turnover, and prevents costly claim denials.	• Patients & Families: Receive clearer goals and faster progress tracking. • Insurance Payers (Medicare / Medicaid): Require objective, measurable proof of medical necessity. • Professional Bodies (AOTA): Advocates for clinician wellness and standards.

Problem Characteristics: Root Causes of the Documentation Gap

Identified Problem Characteristic (Systemic Gap)	Root Cause & Clinical Impact Description	Impact
1. Manual Assessment Synthesis Bottleneck	Standardized assessment batteries (BOT-2, PDMS-2, MMT) yield raw numerical percentiles. Clinicians must manually perform mathematical lookups in paper manuals and translate numbers into prose (by hand) without automation.	Repetitive work
2. Non-Standardized Goal Phrasing	Nothing in UI flags for missing COAST elements (such as explicit assistance levels or measurable environmental conditions). This triggers insurance claim rejections and audit delays.	Missing vital patient information, insurance claim rejections
3. Rigid EHR Documentation Systems	Current clinical software (e.g., WebPT, Epic) provides either rigid drop-down menus that fail to capture nuanced patient needs, or completely blank text boxes that force manual typing from scratch.	Non specific prompts/bad UI
4. Productivity Metric Misalignment	Productivity is measured based on billable patient treatment minutes. Drafting individual treatment goals is de-incentivized since this takes time away from the billable patient treatment minutes.	Less time for patient needs

Current Process Flow: Day In The Life



Core Takeaway: Goal-writing is a repetitive, non-billable cognitive overhead that pulls clinicians away from direct patient care. OT GoalAssist targets steps within the COAST timeline *(yellow highlight)* for intelligent automation.

Proposed Solution: OT GoalAssist Architecture

The Formal Solution Statement:

"OT GoalAssist is an intelligent web application designed for occupational therapy clinicians that accelerates the formulation of individualized, compliant treatment plans. By ingesting standardized clinical assessment results and patient functional priorities, OT GoalAssist leverages a rule-based clinical engine, relational data mapping, and language synthesis to produce editable, COAST-compliant goal suggestions with automated omission detection, preserving clinician autonomy while eliminating repetitive administrative overhead."

Clinician-in-the-Loop **AUTONOMOUS OVERSIGHT**	**COAST Standardized** **5-PART SYNTAX ENGINE**	**Assessment Grounded** **RELATIONAL INGESTION**
Preserves full professional autonomy. The system provides structured recommendations that the licensed clinician inspects, edits, and authorizes under their clinical license.	Enforces complete five-part COAST syntax (Client, Occupation, Assist Level, Specific Condition, Timeframe) via schema validation rules, eliminating omission errors and insurance claim denials.	Directly converts standardized test inputs (BOT-2, PDMS-2, MMT) into functional goal statements, eliminating manual paper scoring manual lookups and mathematical conversion errors.

Solution Characteristics Aligned to Problem Gaps

Problem Characteristic	Solution Characteristic	Solution
Manual Data Synthesis Bottleneck	**Automated Ingestion & Score Normalization**	Interactive quick-entry forms for standardized tests (BOT-2, PDMS-2, MMT). Automatically correlates raw scores, percentile ranks, and age equivalents to identify deficit domains without paper manual lookups.
Non-Standardized Goal Phrasing	**COAST Validation & Omission Guardrails**	Deterministic 5-part syntax parser inspects generated and edited text. Flags missing assistance levels, unquantified conditions, or vague milestones before finalizing, preventing payer audit denials.
Rigid EHR Documentation Systems	**Modular Clinician UI & Universal Clipboard Export**	Split-screen review dashboard allows instant line-by-line clinician edits. One-click universal clipboard bridge exports clean formatted text directly into WebPT, Epic, or SimplePractice without costly API lock-in.
Productivity Metric Misalignment	**4-Tier Accelerated Synthesis Pipeline**	Reduces care plan formulation from 45 minutes down to 10 minutes. Grounded prompt templates synthesize compliant initial drafts in seconds, returning valuable time to direct billable patient therapy.

What OT GoalAssist Will Do

Feature Capability	What the Clinician Experiences & System Output
Structured Clinical Assessment Entry	Provides rapid-entry forms for standardized motor, sensory, and strength assessment scores (BOT-2, PDMS-2, MMT) alongside patient functional priorities.
Rule-Based COAST Goal Formulation	Translates numerical score deficits into clear, individualized 5-part COAST sentences using deterministic clinical phrasing rules.
Real-Time Omission Guardrails	Automatically highlights missing parameters (such as unspecified assistance levels or missing milestones) before care plans can be finalized.
Human-in-the-Loop Clinical Control	Gives clinicians full in-line editing controls, allowing real-time wording adjustments, goal deletion, or alternative phrase generation.
Universal EHR Clipboard Export	Formats approved goals into clean text blocks ready for instant copy-pasting into any existing clinical EHR system (WebPT, Epic, SimplePractice).

What OT GoalAssist Will NOT Do

OT GoalAssist will NOT	Why?
Replace any jobs	OT GoalAssist is simply a tool to help therapists do their job more efficiently by having a dedicated goal and note taker to track the progress and important contexts of individual patients. OT GoalAssist will only assist in keeping track with patients, not treating them. Therapists will still need to authorize goals with their license as liability.
Provide any medical advice or tell therapists how to do their job	OT GoalAssist is a tool to help a therapist keep track of their patients progress, situation and other important contexts or changes regarding them. It will not give therapists any information about how to treat patients or proceed with treatment.
Directly keep any patient information	OT GoalAssist is not a tool for patients and will not keep any information regarding them besides the relevant information that is explicitly provided to the therapist.

Technical Feasibility Snapshot: Architecture Overview

Tier 1: Client Layer	Tier 2: Logic & Rules	Tier 3: Language Synthesis	Tier 4: Secure Data
Web Client (HTML5 / CSS / JavaScript)	**Clinical Rule Engine (Python / Node.js)**	**Grounded Language Layer (Prompt Guardrails)**	**Relational & Session Store (Zero Persistent PHI)**
• Quick-input forms for assessment batteries. • Real-time error highlighting. • Split-screen review interface. • One-click export to clipboard.	• Enforces 5-part COAST syntax. • Validates assistance levels. • Checks milestone timeframes. • Flags incomplete goal fields.	• Formats rough inputs into clinical prose. • Strictly grounded in therapist entries. • Prevents hallucinated medical facts. • Generates alternative phrasings.	• Session-anonymized processing. • No patient identifiers stored. • HIPAA compliant architecture. • Encrypted transmission protocols.

Major Functional Components Diagram (MFCD)

1. User & External Interfaces	2. OT GoalAssist Software Engine	3. Data & Storage Subsystems
Clinical Users & External Systems	**Core Processing Pipeline**	**Secure Database Layer**
• Licensed OT Clinician: Inputs assessment scores, functional challenges, and patient priorities. • Standardized Test Batteries: Raw scores and percentiles from BOT-2, PDMS-2, and MMT. • Target Clinical EHR: Universal clipboard recipient (WebPT, Epic, ClinicSource).	• Assessment Ingestion Parser: Calculates age equivalents and deficit domains. • Deterministic COAST Rule Parser: Validates Client, Occupation, Assist, Condition, Time. • Grounded Language Synthesis Engine: Formats raw notes into cohesive clinical goal sentences. • Split-Screen Clinician Review Interface: Live inline editing, approval, and clipboard export.	• Relational Database: Stores assessment records, deficit benchmarks, and goal templates. • Zero-PHI Ephemeral Storage: Patient data is de-identified client-side; no persistent PII/MRNs stored. • Secure HTTPS / TLS 1.3: Encrypted transmission protecting all therapist interactions.

Relational Database Schema & Data Flow (ERD)

TABLE: Clinicians	TABLE: Patient_Sessions	TABLE: Assessment_Records	TABLE: COAST_Goals
PK: clinician_id (UUID)	PK: session_id (UUID)	PK: record_id (UUID)	PK: goal_id (UUID)
• npi_number (VARCHAR) • license_state (CHAR) • full_name (VARCHAR) • clinic_name (VARCHAR) • auth_hash (TEXT) Logs therapist licensing credentials for care plan authorization.	• patient_token (HASH) • clinician_id (FK) • age_months (INT) • care_setting (ENUM) • created_at (TIMESTAMP) Zero-PHI session. Links evaluations to randomized pseudonymous tokens.	• session_id (FK) • test_battery (ENUM) • composite_score (INT) • percentile_rank (FLOAT) • deficit_domain (VARCHAR) Stores standardized assessment scores (BOT-2, PDMS-2, MMT).	• session_id (FK) • client_c (TEXT) • occupation_o (TEXT) • assist_a (VARCHAR) • condition_s (TEXT) • timeframe_t (VARCHAR) • clinician_approved (BOOL) Stores structured 5-part COAST syntax components.

Core Database Architecture Principles: (1) Client-Side De-Identification strips patient names and MRNs before database transmission. (2) Ephemeral Session Records purge automatically upon export. (3) Strict Access Controls restrict data retrieval solely to the authenticated licensed therapist.

Development Tools & Implementation Environment

Architecture Layer	Selected Technology	Clinical & Operational Rationale
Frontend Client	**HTML5, Modern CSS, JavaScript**	Clean interactive forms, real-time error validation styling, universal clipboard API support, and zero complex build overhead.
Application Backend & Logic	**Python / Node.js Runtime**	Implements deterministic COAST syntax parser, standardized assessment score calculators, and API request routing.
Relational Database	**SQL Relational Database**	Structured tables for assessment score normalization, deficit lookup tables, and session-isolated goal history.
Language Synthesis Layer	**Language Model API + Prompt Guardrails**	Converts structured clinical notes into cohesive professional prose with strict negative prompt constraints preventing hallucinations.
Version Control & Collaboration	**Git, GitHub, GitHub Pages**	Centralized repository management, branch pull-request workflows, and public project portal hosting.

Technical & Security Risk Mitigation Matrix

Risk Domain & Description	Potential Impact	Severity / Prob	Technical & Operational Mitigation Strategy
T1: Fact Hallucination	Language model invents patient deficits or unrealistic milestones.	**HIGH / LOW**	Strict prompt bounding anchored exclusively to entered test scores; rule engine cross-checks outputs; mandatory therapist sign-off before export.
S1: PHI & HIPAA Exposure	Leakage of identifiable patient health information.	**CRITICAL / VERY LOW**	Zero Persistent PHI architecture. Client-side scrubbing of direct identifiers; ephemeral session tokens; TLS 1.3 encrypted transmission.
O1: Payer Claim Denial	Goal phrasing lacks required elements, triggering insurance audit rejections.	**MED / LOW**	Embedded COAST syntax parser checks for all 5 required elements (Client, Occupation, Assist, Condition, Time) prior to export.
I1: EHR Incompatibility	Target clinic EHR software blocks direct API integration.	**MED / VERY LOW**	Universal clipboard export bridge formats finalized goals into clean text ready for instant 1-click paste into WebPT, Epic, or ClinicSource.

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Competitive Analysis: Market Gaps & True Innovation

Features	OT GoalAssist	WebPT / ClinicSource	SimplePractice	Generic LLMs	Word Documents
Standardized Test Synthesis	✔	◐	✘	◐, Risky	✘
C.O.A.S.T. Syntax Enforced	✔	✘	✘	◐, Unreliable	◐, Manual Review Only
Omission Guardrails	✔	◐, AI assist in certain aspects	◐, Only allows carry-over	✘, can fill gaps itself	✘
Clinician Stays in the Loop	✔	✔	✔	✘	✔
Low Cost	✔	✘	✘	✔	✔
Goal Generation	✔	✔	✔	◐, Unrealistic Goals	✘
Privacy Safeguards	✔	✔	✔	✘	✘

Core Differentiator: Aggregation is not innovation. Existing tools either offer rigid drop-downs or unconstrained text. OT GoalAssist is the first tool to directly translate standardized evaluation numbers into validated COAST goals with clinical omission guardrails.

Yes ✔ | Partial ◐ | No ✘

Clinical & Technical Glossary

Acronym / Clinical Term	Clinical Definition & Operational Context
ADL (Activities of Daily Living)	Fundamental self-care tasks required for daily independent living (dressing, eating, bathing, functional mobility).
AOTA	American Occupational Therapy Association; the primary national professional association representing occupational therapy clinicians.
BOT-2	Bruininks-Oseretsky Test of Motor Proficiency (2nd ed.); standardized clinical assessment battery measuring fine and gross motor skills.
COAST	Clinical goal documentation framework: Client, Occupation, Assist Level, Specific Conditions, and Time Frame.
COTA	Certified Occupational Therapy Assistant; works under registered occupational therapist supervision to deliver clinical interventions.
EHR	Electronic Health Record software platform utilized by clinical facilities to store patient charts and billing encounters.
MMT	Manual Muscle Testing; standardized 0-to-5 clinical scale used by therapists to grade patient muscular strength.
OTR	Occupational Therapist, Registered; licensed clinician qualified to conduct diagnostic evaluations and sign care plans.
PDMS-2	Peabody Developmental Motor Scales (2nd ed.); standardized assessment measuring early childhood gross and fine motor abilities.
POC (Plan of Care)	Formal clinical treatment roadmap detailing patient diagnoses, frequency, duration, and measurable functional goals.

References

Ref	Complete Academic / Regulatory Citation
[1]	U.S. Department of Health and Human Services. Health Insurance Portability and Accountability Act (HIPAA) Security Rule Standards.
[2]	Medicare Benefit Policy Manual (Ch. 15) Covered Medical and Other Health Services
[3]	Bruininks, R. H., & Bruininks, B. D. (2005). Bruininks-Oseretsky Test of Motor Proficiency (2nd ed.) (BOT-2).
[4]	American Occupational Therapy Association (AOTA) (2020) Occupational Therapy Code of Ethics
[5]	National Library of Medicine (2025) Usker Naqvi; Konstantinos Margetis; Andrew L. Sherman: Muscle Strength Grading
[6]	National Library of Medicine (2024) Peabody Developmental Motor Scales (2nd Ed) Anna Chalupka Borowska, Magdalena Sobieska
[7]	Taylor & Francis Shaping the Goal Setting Process in OT: Verna G. Eschenfelder (2009)